

Problem

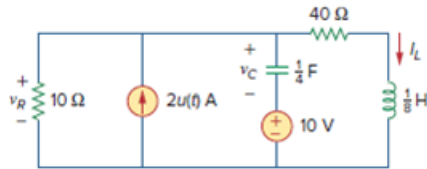
Refer to the circuit shown in Fig. 8.64. Calculate:

(a) $i_L(0^+)$, $v_C(0^+)$, and $v_R(0^+)$,

(b) $di_L(0^+)/dt$, $dv_C(0^+)/dt$, and $dv_R(0^+)/dt$,

(c) $i_L(\infty)$, $v_C(\infty)$, and $v_R(\infty)$.

Figure 8.64:



Step-by-step solution

Step 1 of 7

(a)

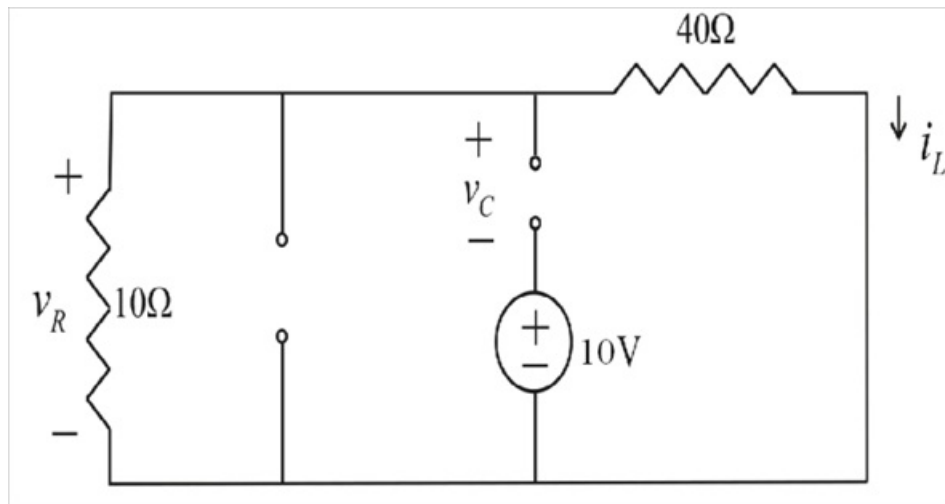
Refer to the circuit diagram in Figure 8.64 in the textbook.

For $t < 0$, current source, $2u(t) = 0$,

At $t = 0^-$, since the circuit has reached steady state, the inductor is acting as a short circuit and the capacitor as an open circuit.

Step 2 of 7

At $t = 0^-$, the circuit becomes:



Therefore, $i_L(0^-) = 0A$, $v_C(0^-) = -10V$, $v_R(0^-) = 0V$

[Comments \(5\)](#)



Anonymous

why is the CURRENT source now acting as a open circuit? shouldn't a current source act a short circuit?



Anonymous

Before $t=0$ the current source has zero current going through it, so it is like an open circuit



Anonymous

I believe it's a matter of inserting the current source at $t > 0$. It shows towards then end what happens in the steady state condiiton



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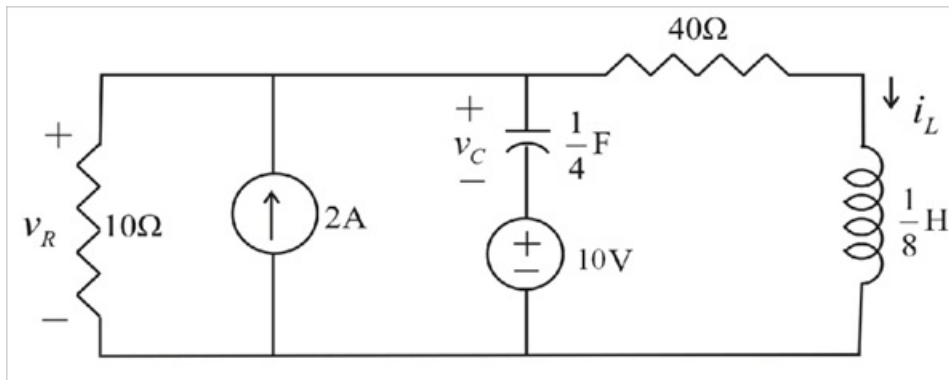
refer to book as practice problem 8.1



Anonymous

the current source looks open but in other examples in problems they achieve the same effect by inserting a switch to cut out the amp source. So you could redraw the above with an open switch just above the current source.

For $t > 0$, $2u(t) = 2$, the circuit becomes:



Since the inductor current and capacitor voltage cannot change abruptly.

$$i_L(0^+) = 0\text{A}$$

$$v_C(0^+) = -10\text{V}$$

$$v_R(0^+) = 0\text{V} \quad \text{Since } v_C \text{ and } 10\text{ V voltage source is parallel to } 10\ \Omega$$

Step 4 of 7

(b)

Voltage across the inductor is expressed as,

$$L \frac{di_L}{dt} = v_L$$

Differentiate both sides.

$$\frac{di_L(0^+)}{dt} = \frac{v_L(0^+)}{L}$$

Since current through the inductor cannot change instantaneously.

$$v_L(0^+) = 0$$

$$\frac{di_L(0^+)}{dt} = 0$$

Step 5 of 7

Current flow through the capacitor is expressed as,

$$\text{Similarly } C \frac{dv_C}{dt} = i_C$$

Applying KCL at node v_R .

$$2 = i_R(0^+) + i_C(0^+) + i_L(0^+)$$

Substitute $C \frac{dv_C(0^+)}{dt}$ for $i_C(0^+)$, 0 for $i_R(0^+)$ and 0 for $i_L(0^+)$ in the equation.

$$\begin{aligned} \frac{dv_C(0^+)}{dt} &= \frac{i_C(0^+)}{C} \\ &= \frac{2}{\left(\frac{1}{4}\right)} \\ &= 8 \end{aligned}$$

$$\frac{dv_C(0^+)}{dt} = 8\text{V/s}$$

Step 6 of 7

Apply KVL around voltage source, capacitor and $10\ \Omega$ resistor.

$$10 + v_C(0^+) - v_R(0^+) = 0$$

Differentiate both sides.

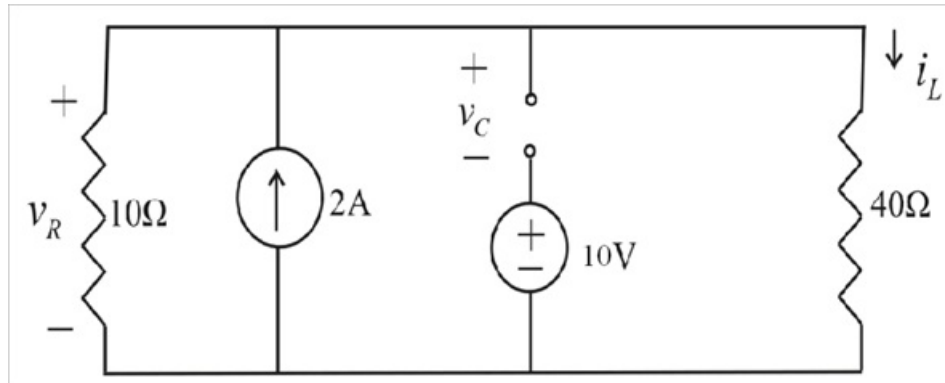
$$\frac{dv_R(0^+)}{dt} = \frac{dv_C(0^+)}{dt}$$

$$\boxed{\frac{dv_R(0^+)}{dt} = 8\text{V/s}}$$

Step 7 of 7

(c)

As $t \rightarrow \infty$, the circuit reaches steady state, therefore the circuit is



Find current flow through the inductor using current division rule.

$$i_L(\infty) = 2 \times \frac{10}{10 + 40}$$

$$\boxed{i_L(\infty) = 0.4\text{A}}$$

Voltage across the resistor is,

$$v_R(\infty) = 10 \times i_L(\infty)$$

$$= 10 \times (2 - i_L(\infty))$$

$$= 10 \times (2 - 0.4)$$

$$\boxed{v_R(\infty) = 16\text{V}}$$

Apply KVL around voltage source, capacitor and $10\ \Omega$ resistor.

$$v_C(\infty) = v_R(\infty) - 10$$

$$= 16 - 10$$

$$\boxed{v_C(\infty) = 6\text{V}}$$